Business Data Analysis using OLAP and Apache Superset

BY-

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OLAP Operations

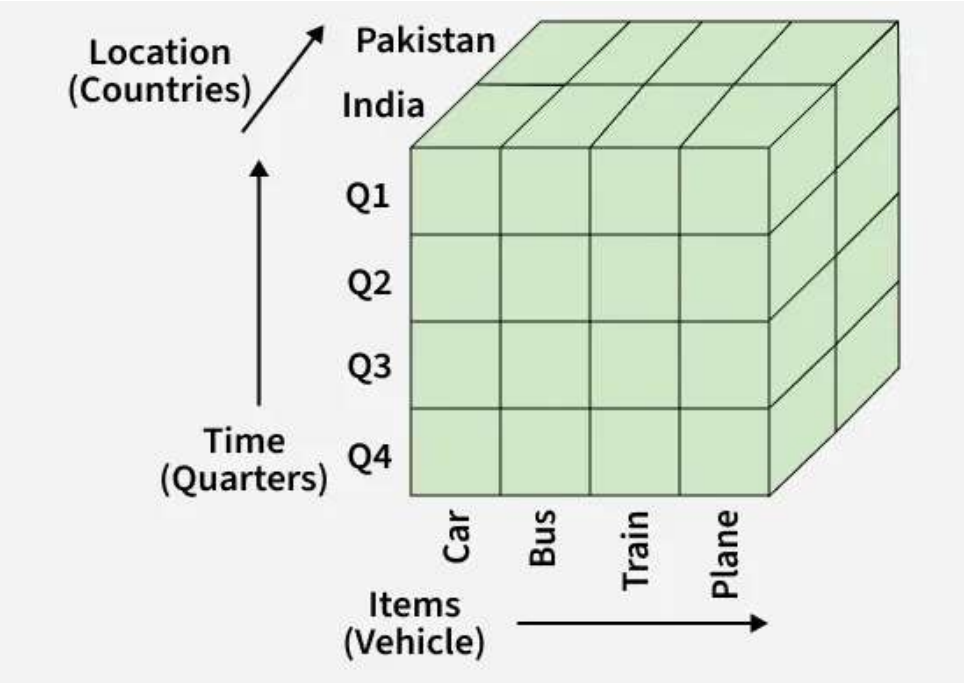
- OLAP (Online Analytical Processing) is a software **technology** that enables users to analyze data from **multiple database systems** simultaneously by the help of **ETL Process (Combine all data)**

  Extract → Take data from all sources

  Transform → Clean & convert into same format

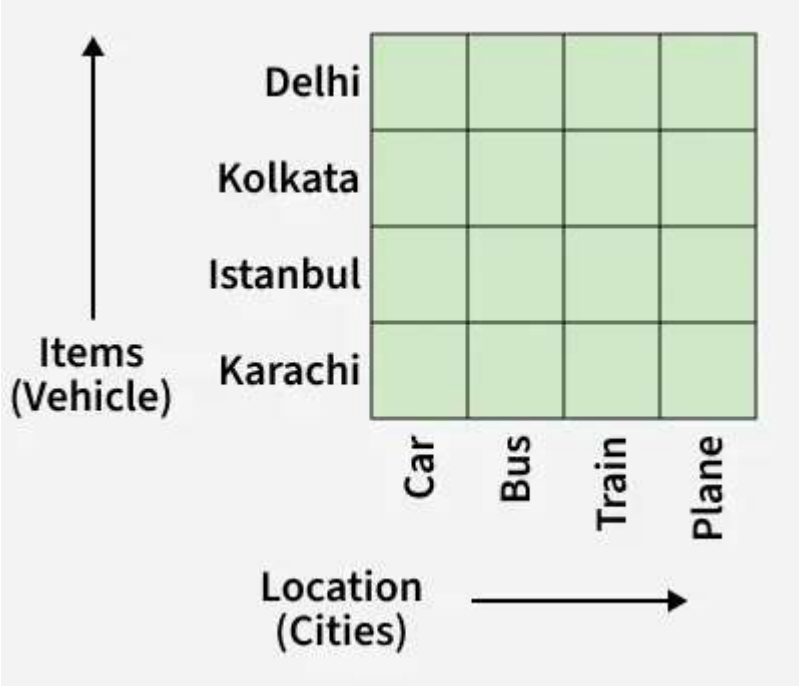
  Load → Store in one place

- It is based on a **multidimensional data model**, where data is represented in the form of cubes, also known as hyper-cubes.
- Each cube consists of dimensions (e.g., Location, Time, Product) and measures (e.g., Sales, Profit).



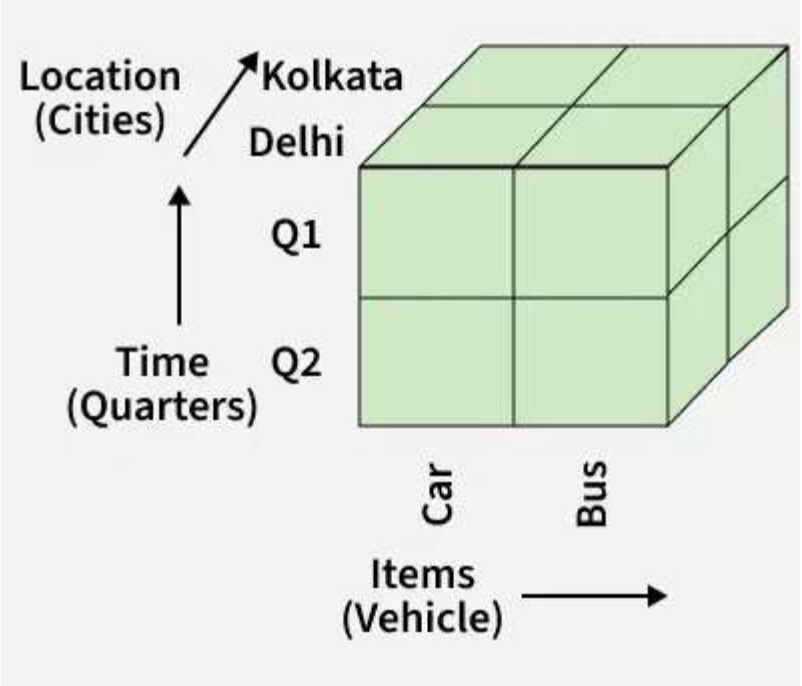
OLAP Operations

- **Slice**
- The Slice operation selects **a single dimension from the cube**, creating a new **sub-cube** with reduced dimensionality.
- It helps focus on a specific data slice for analysis.
- ***Example:*** *Selecting Time = "Q1" to analyze sales across all products and regions for the first quarter.*



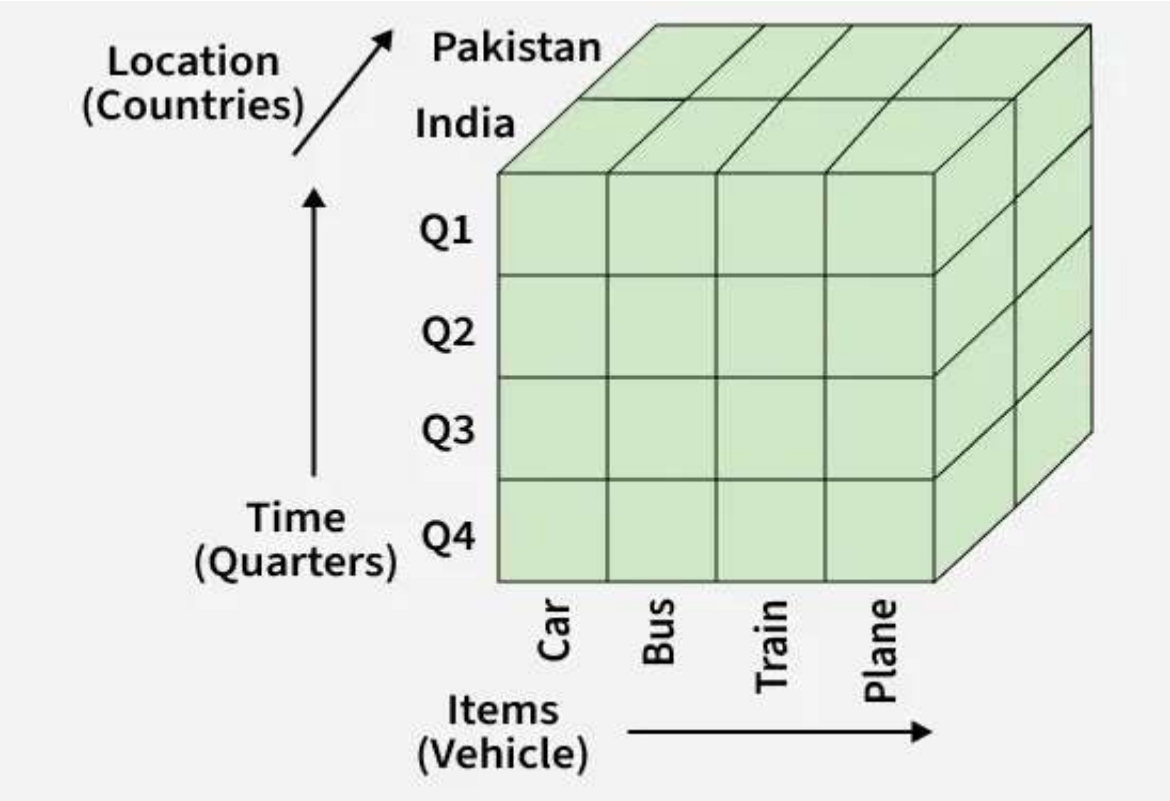
OLAP Operations

- **Dice**
- The Dice operation selects data from the cube by applying **filters on two or more dimensions** to form a sub-cube.
- This results in a smaller cube focused on these **specific dimensions**.
- ***Example:*** *Selecting: Location = "Delhi" or "Kolkata", Time = "Q1" or "Q2" & Item = "Car" or "Bus"*



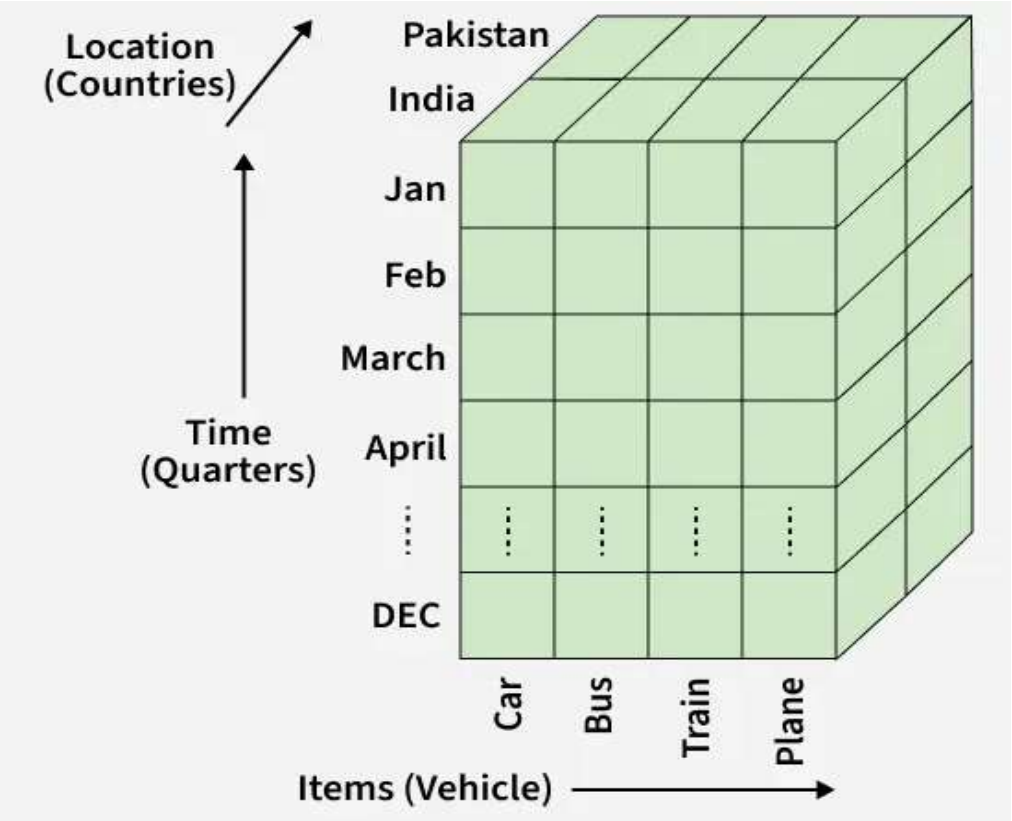
OLAP Operations

- **Roll Up(Summarize)**
- The Roll Up operation is the opposite of Drill Down.
- It aggregates or summarizes data to provide a **higher-level overview**.
- **How it works:** Climbing up in the concept hierarchy **& reducing the number of dimensions**.
- ***Example:*** *Aggregating sales data from City -> Country or from Month -> Quarter.*



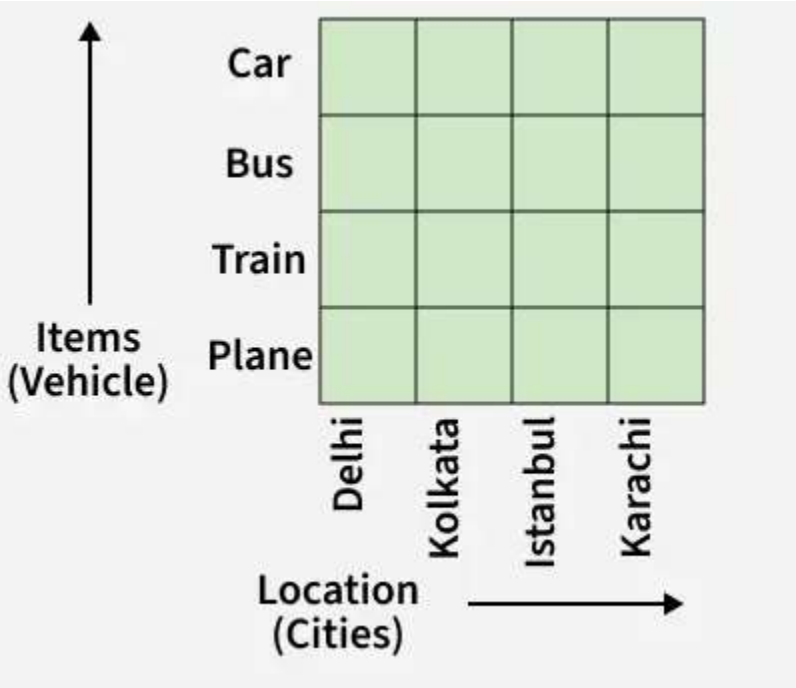
OLAP Operations

- **Drill Down**
- The Drill Down operation provides a more **detailed view** of the data.
- It moves from a summary level to a lower level in the concept hierarchy (for example, from **Year -> Quarter -> Month).**
- **How it works:** Moving down in the hierarchy & adding new dimensions for more granularity.
- ***Example:*** *Viewing sales data for 2024 -> Q1 -> January instead of only yearly totals.*



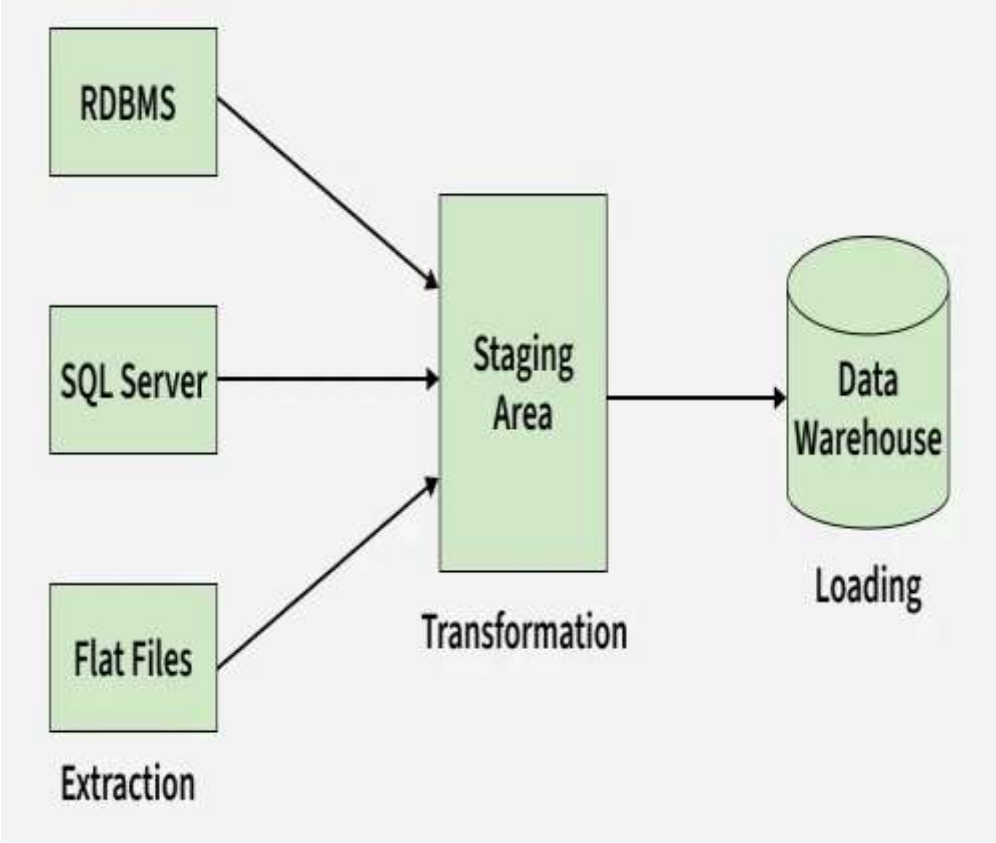
OLAP Operations

- **Pivot (Rotation)**
- The Pivot operation (also known as Rotation) reorients the cube to provide a different view of the data.
- It helps users visualize data from different perspectives by rotating rows and columns.
- ***Example:*** *Swapping the Time and Location axes to compare sales by quarter across different regions.*



Data Loading (ETL Process)

- ETL (Extract, Transform, Load) is a key process in data warehousing that prepares data for analysis. It involves:
- **E**xtracting data from multiple sources
- Transforming it into a consistent format
- Loading it into a central data warehouse or data lake
- ***Note:*** *ETL helps businesses unify and clean data, making it reliable and ready for analysis. It improves data quality, security and accessibility, enabling better insights and faster decision-making in a world of diverse data sources.*

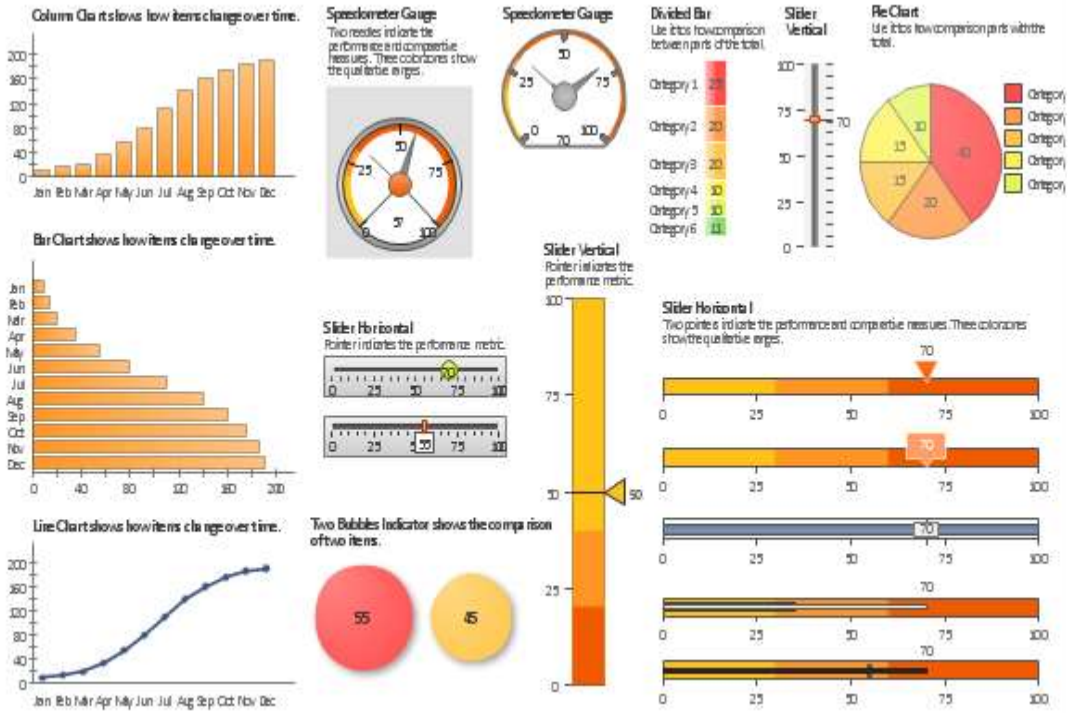


Apache Superset

- Apache Superset is an **open-source data visualization and business intelligence (BI) tool** used to:
- Create dashboards
- Analyze large datasets
- Run SQL queries
- Generate charts and reports

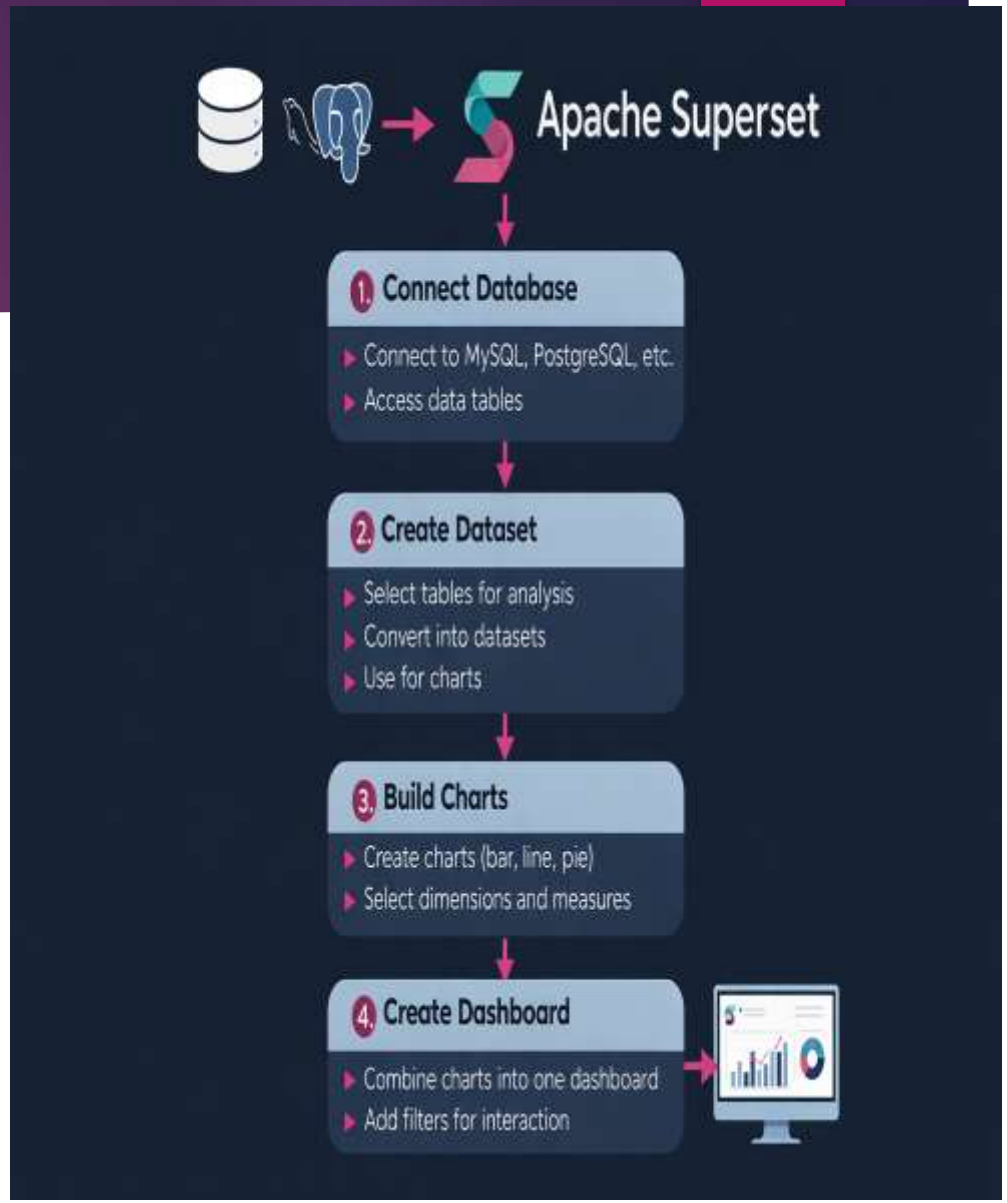
It is similar to tools like:

- Tableau
- Power BI



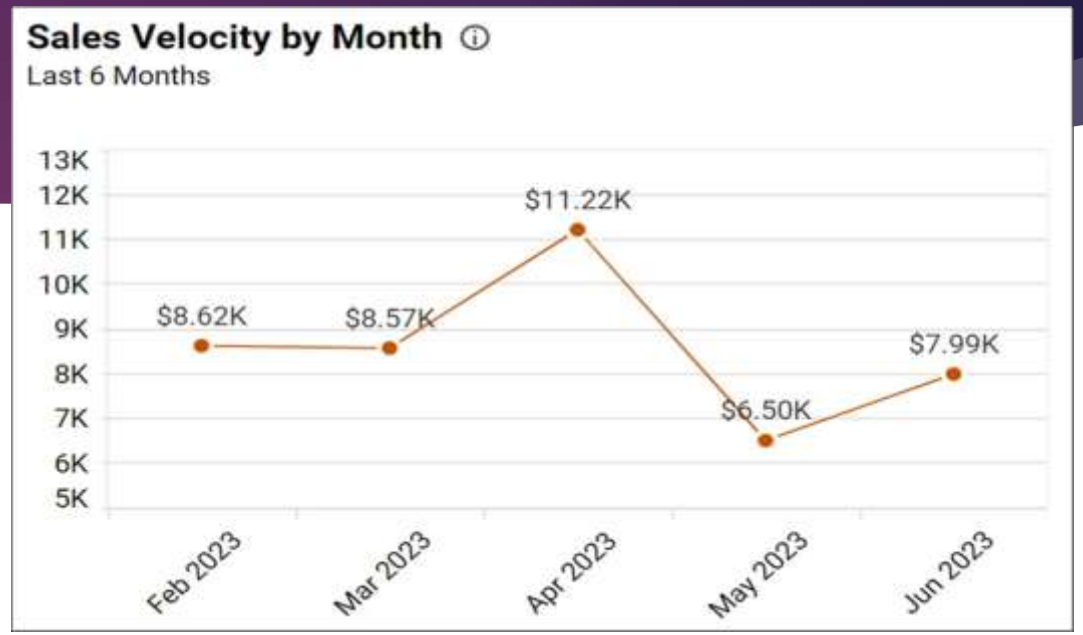
Working with Superset

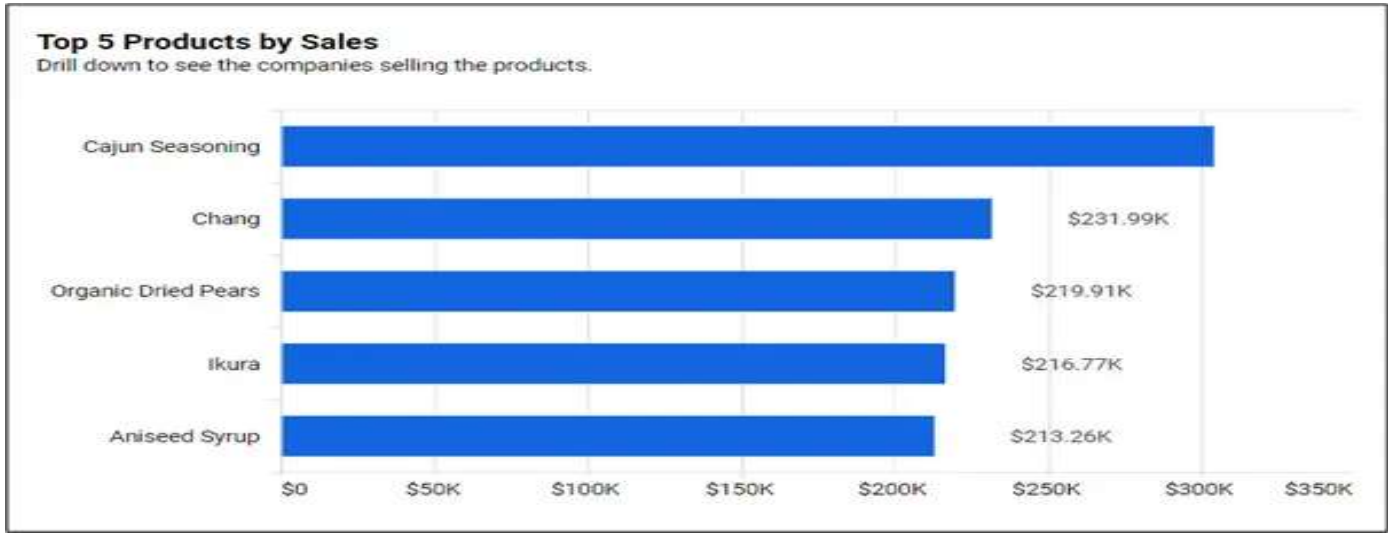
- **1. Connect Database**
- Connect Superset to database (MySQL, PostgreSQL, etc.)
- Access data tables
- **2. Create Dataset**
- Select tables for analysis
- Convert into datasets
- Use for charts
- **3. Build Charts**
- Create charts (bar, line, pie)
- Select dimensions and measures
- **4. Create Dashboard**
- Combine charts into one dashboard
- Add filters for interaction



Practical Example – Sales Data Analysis

- Analyze sales dataset (product, date, region, revenue)
- Identify top-selling products using bar chart
- Analyze monthly sales trends using line chart
- Show category or region share using pie chart
- Create dashboard to combine all charts





Product Sales Dashboard

Auto date range
This Week

Services
All

Posts
All

Total Revenue

$86.1

↑ 13%

vs previous 7 days

Sales Conversion Rate

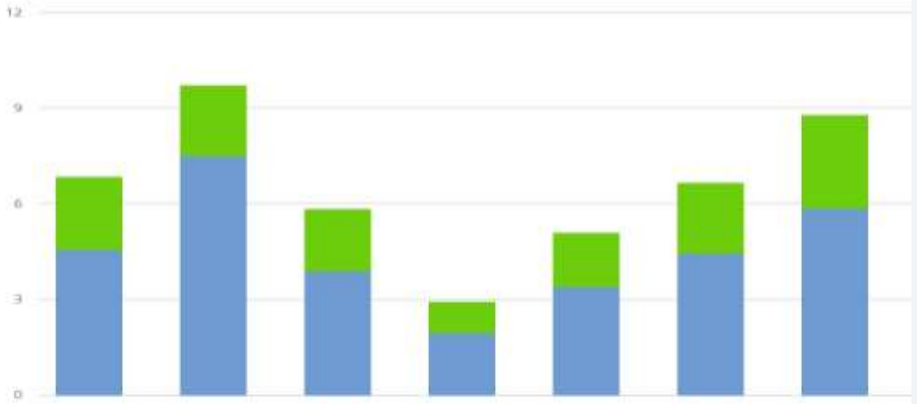
26%

8%

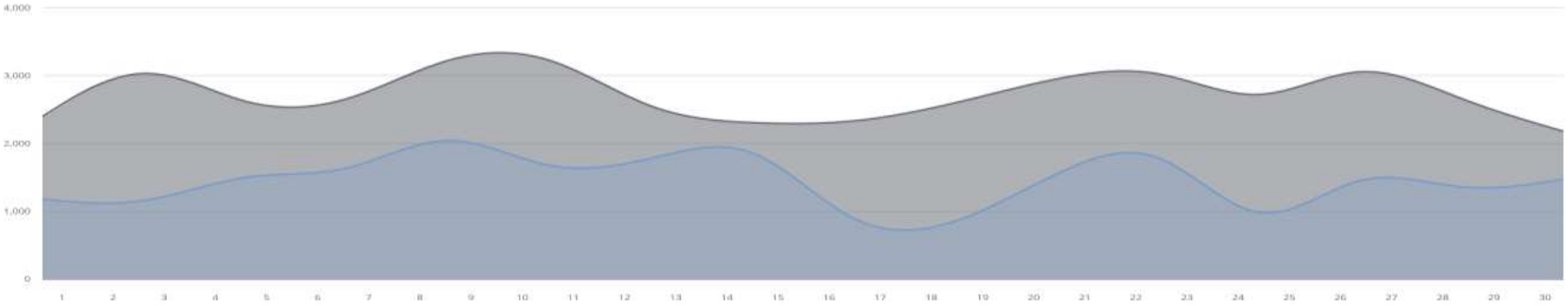
vs previous 7 days

Sales Over Time

Avg Sales Value



Sales Conversion Rate



Thank You